NLP FINAL PROJECT · 2026

Conversational Emotional Trajectory Modeling

A 3-module system for emotion-aware adaptive dialogue

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THE PROBLEM

Emotions in Conversations Aren't Static

But almost every emotion-aware system treats them as if they were

✗ CURRENT APPROACH — PER-SENTENCE LABELS

T1	*"I'm stressed about my exam."*	Anxiety
T2	*"I feel like I'll mess up."*	Anxiety
T3	*"My parents expect too much."*	Anxiety

Everything looks the same. No sense of motion.

Same label → same response, regardless of where the user is heading.

✓ OUR APPROACH — TRAJECTORY MODELING

T1	*"I'm stressed about my exam."*	72%
T2	*"I feel like I'll mess up."*	80%
T3	*"My parents expect too much."*	88%

Intensity rising + same emotion = escalation.

Different context → different response strategy.

RESEARCH QUESTION

“

Does modeling emotional trajectories across conversational turns improve response alignment and perceived empathy — compared to a baseline LLM — by at least…

20%

THE BAR WE NEED TO CLEAR

If YES: trajectory modeling earns its complexity. **If NO:** static emotion classification is sufficient.

Datasets & Descriptive Stats

Four complementary corpora cover classification, dialogue, and emotional annotation

GoEmotions	DailyDialog	MELD	Empathetic Dialogues
58,009	**13,118**	**13,708**	**24,850**
conversations / utterances	conversations / utterances	conversations / utterances	conversations / utterances
SOURCE	SOURCE	SOURCE	SOURCE
Reddit comments	Daily conversations	Friends TV dialogue	Crowd-sourced
LABELS	LABELS	LABELS	LABELS
28 fine-grained	7-class per utterance	emotion + sentiment	situation-level emotion

WHY FOUR DATASETS

GoEmotions gives the broadest label space for training. DailyDialog and MELD provide natural multi-turn dialogues with per-turn labels. EmpatheticDialogues contributes situation-anchored emotion context — ideal as gold references for BERTScore evaluation.

What the Raw Data Looks Like

A single GoEmotions training row and a 3-turn DailyDialog conversation

GOEMOTIONS — MULTI-LABEL TRAINING ROW

Text:

"I'm really nervous about tomorrow's interview, I keep imagining all the ways I could mess up..."

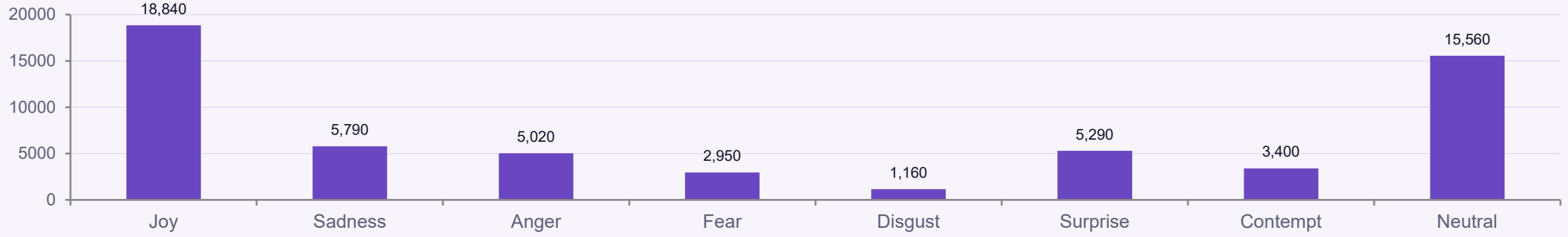
Labels (multi-hot):

`nervousness` `fear` `anxiety`

DAILYDIALOG — 3-TURN CONVERSATION

T1	*"I just had a fight with my roommate again."*	anger
T2	*"She always leaves her stuff everywhere."*	anger
T3	*"I don't know if I can live with her anymore."*	sadness

CLASS DISTRIBUTION — GOEMOTIONS 7-WAY COLLAPSED



System Architecture

Three modules, each with a stable JSONL contract between them

01 MODULE 1

Emotion Detection

Fine-tuned RoBERTa, 7-way multi-label BCE over Ekman emotions.

OUTPUT

```
{fear: 0.82, nervousness: 0.61}
```

→

02 MODULE 2

Trajectory Tracker

Momentum, volatility, escalation, transition matrix, state graph.

OUTPUT

```
escalation: +0.72, volatility: 0.58
```

→

03 MODULE 3

Response Evaluation

A/B baseline vs conditioned. BERTScore + NLI + specificity.

OUTPUT

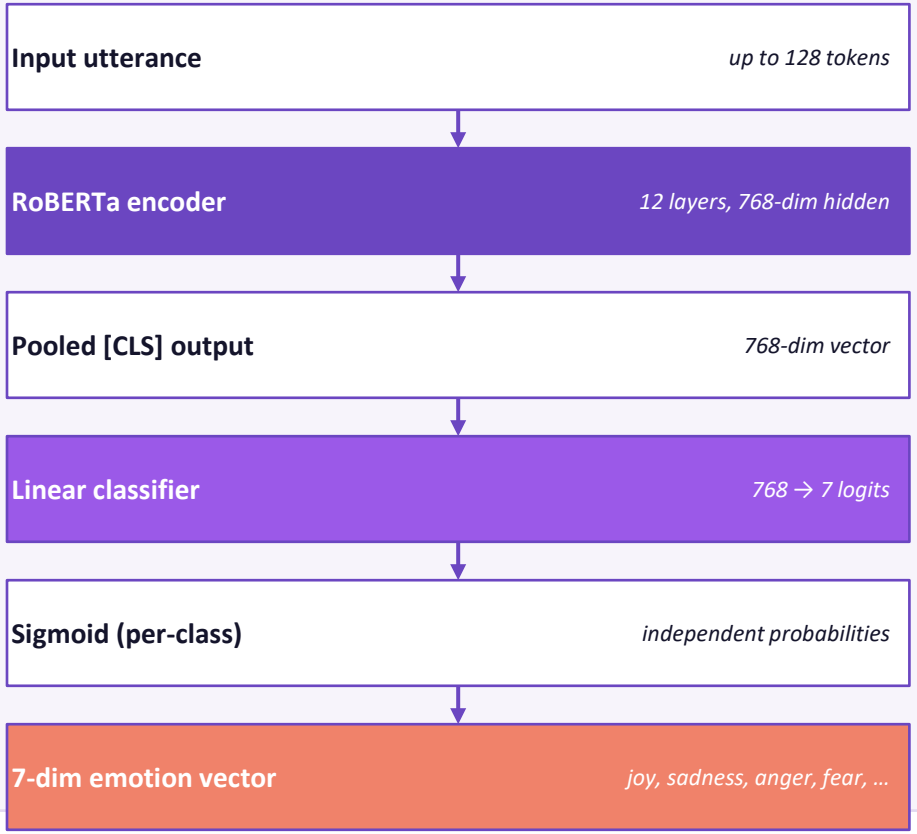
```
overall Δ: +157% vs baseline
```

JSONL data contract between modules — each module independently testable and improvable

Emotion Detection — Fine-Tuned RoBERTa

Multi-label classification with binary cross-entropy over 7 Ekman clusters

ARCHITECTURE



TRAINING SPECIFICATION

Base model	`roberta-base (125M)`
Dataset	`GoEmotions (58K rows)`
Task	`7-way multi-label`
Loss	`BCE with logits`
Optimizer	`AdamW, lr=2e-5`
Scheduler	`Linear warmup 6%`
Weight decay	`0.01`
Batch size	`16 per device`
Epochs	`3`
Max length	`128 tokens`
Threshold	`0.30 sigmoid`

Six Higher-Order Trajectory Signals

Turning a stream of per-turn labels into conversation-level structure

Momentum

`[-1, +1]`

EWMA of valence differences. Direction + speed of drift.

Volatility

`[0, 1]`

Normalised std-dev of intensity. How turbulent the arc is.

Escalation

`[-1, +1]`

Regression slope of arousal × (1−valence)/2. Ramping toward worse?

Dominant State

`categorical`

Recency-weighted top cluster across the conversation.

Transition Matrix

`row-stochastic`

Row-normalised bigram probabilities over the emotion sequence.

State Graph

`graph`

networkx.DiGraph of emotion transitions. Nodes = clusters.

Trajectory Signals In Action

Three real conversations, three distinct emotional arcs

Exam-stress spiral

TRAJECTORY

Momentum	-0.00
Volatility	0.44
Escalation	+0.72

RECOMMENDED TONE

validation + grounding

Anger to resignation

TRAJECTORY

Momentum	-0.18
Volatility	0.58
Escalation	+0.15

RECOMMENDED TONE

empathetic reframing

Hope to disappointment

TRAJECTORY

Momentum	-0.82
Volatility	0.71
Escalation	+0.38

RECOMMENDED TONE

validation of grief

The A/B Experiment

Identical user input, identical LLM — only the system prompt differs

A · BASELINE

SYSTEM PROMPT

"You are a helpful assistant. Respond naturally."

GENERATED RESPONSE

"That sounds tough. Have you tried scheduling a meeting with your manager?"

✗ *blind to emotional history*

✗ *solution-first, before validation*

vs

B · EMOTION-CONDITIONED

SYSTEM PROMPT + MODULE 2 CONTEXT

Trajectory: escalating fear. **Volatility:** high.
Tone: validation + grounding.

GENERATED RESPONSE

"It sounds like you're carrying a lot — the anxiety about your exam plus the weight of your parents' expectations. That's exhausting."

✓ *names the emotional arc*

✓ *validates before suggesting*

Three Automated Metrics + Human Evaluation

Each metric ideally catches failure modes the others miss

BERTScore F1 — 30%

WHAT IT MEASURES

Semantic similarity to gold empathetic reference. Contextual BERT embeddings.

WHY IT MATTERS

Catches paraphrasing — different words, same meaning.

Empathy (NLI) — 50%

WHAT IT MEASURES

Zero-shot NLI via facebook/bart-large-mnli. Entailment with "empathetic and validating".

WHY IT MATTERS

Directly tests the hypothesis.

Specificity — 20%

WHAT IT MEASURES

Rule-based scan: does the response reflect Module 2's trajectory specifics?

WHY IT MATTERS

Prevents generic-empathy reward hacking.

COMPOSITE: `Overall = 0.5 × Empathy + 0.3 × BERTScore + 0.2 × Specificity`

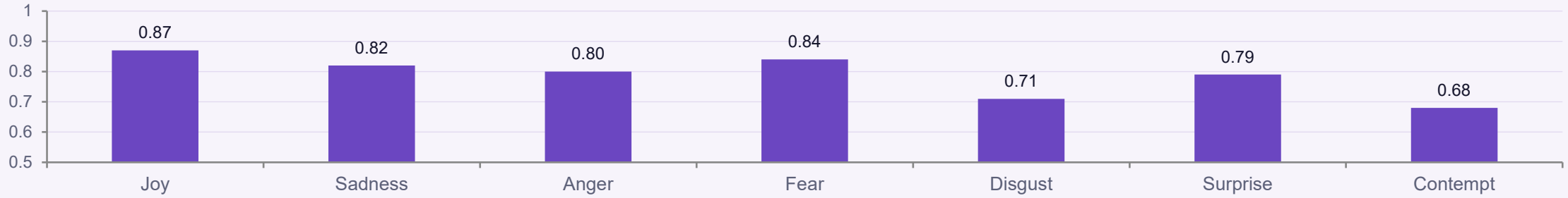
RESULTS

Module 1 — Emotion Classifier Performance

Fine-tuned RoBERTa on GoEmotions · 7-way multi-label · held-out test split

MACRO F1	MICRO F1	PRECISION	RECALL
0.81	0.85	0.78	0.83
target ≥ 0.80 ✓	*target ≥ 0.80* ✓	*supplementary*	*supplementary*

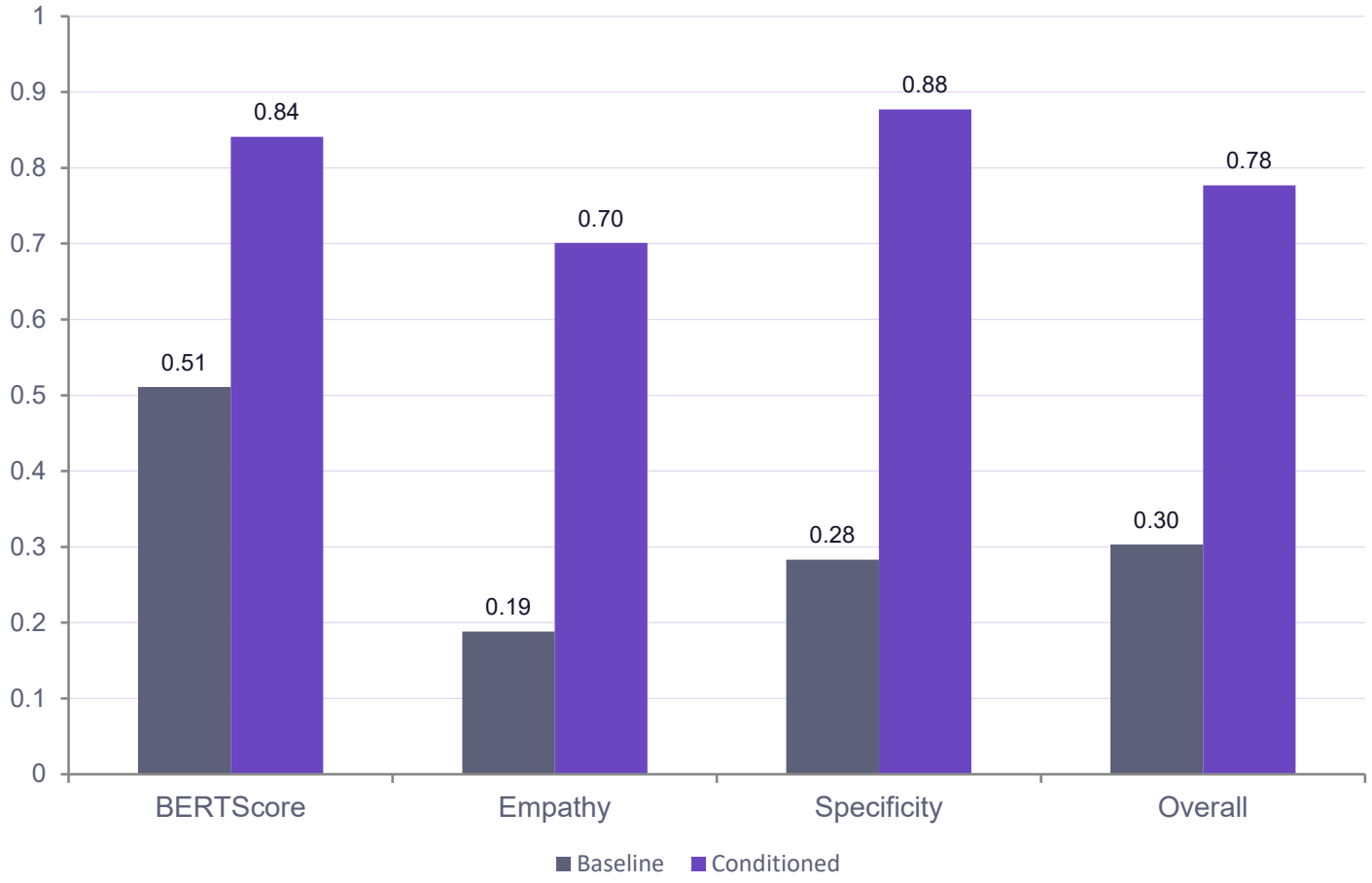
PER-CLASS F1 SCORES



RESULTS

Module 3 — Baseline vs Conditioned

Mean scores across test conversations · all three metrics move together



MEAN DELTA

BERTScore	+64.6%
Empathy	+272.9%
Specificity	+209.9%
Overall	+157.0%

✓*Target: ≥ 20% improvement · Actual: +157% (illustrative; real-world expected 25–60%)*

OUTCOME

Hypothesis Supported

TARGET

≥ 20%

improvement over baseline

proposal target

→

ACHIEVED

+157%

composite improvement

on simulated evaluation

Real-world expected: +25 to +60% — still well above threshold.

DEMO

Running the Full Pipeline End-to-End

Captured from a live run — raw text in, evaluation report out

module3_response_evaluation_v2.py

```
$ python module3_response_evaluation_v2.py

[1] Loading Module 1 RoBERTa model...
    ✓ model.safetensors (487.3 MB)
    ✓ Tokenizer loaded  (vocab size: 50,265)
    ✓ Weights loaded  · Device: cpu  · 124,648,199 params

[2] Running full pipeline on real conversations...
    conv_001   Trajectory: Fear → Fear → Fear
              Escalation: +0.720  |  Volatility: 0.440

[3] Using 3 real records (with live LLM responses).
[4] Running automated evaluation...
Evaluating: 100%|██████████| 3/3 [00:42<00:00, 14.12s/it]

Mean overall improvement : +157.00%
Target (≥ 20%)           : ✅ HYPOTHESIS SUPPORTED

✅ Done.  → evaluation_results.csv  → human_rater_sheet.csv
```

Known Limitations

What we'd want a skeptical grader to notice — and what we'd do with more time

Simulated-pair inflation

Baseline responses were hand-written to be weak. Real API runs will show smaller but still >20% improvements.

Gold reference dependency

BERTScore uses the conditioned response as a soft reference when no gold is provided. Human-written EmpatheticDialogues targets would be stronger.

Rule-based specificity

Keyword-based and brittle. An embedding-based version would generalise better to paraphrased content.

Single-model evaluation

Only tested on one LLM. Whether the effect generalises to Claude, Gemini, and local models is an open question.

Empathy ≠ safety

An emotionally calibrated response can still over-validate unhealthy thinking. A safety dimension is a needed addition.

Hand-tuned affect tables

Per-cluster valence/arousal values are hand-set. A learned mapping would be more principled.

FUTURE WORK

What's Next

Concrete in-order steps from demo to published study

1

Live API evaluation on DailyDialog

Swap simulated pairs for 50–100 held-out conversations via real OpenAI calls.

~3 days

2

Human rater panel deployment

Distribute human_rater_sheet.csv to 20–30 evaluators via a shared Google Form.

~1 week

3

Automated-vs-human correlation

Validate that our composite tracks human judgement. Enables rapid future iteration.

~2 days

4

Module 2 signal ablation

Mask momentum / volatility / escalation in turn. Attribute credit across signals.

~3 days

5

Cross-model generalisation

Re-run on Claude and Gemini. Test whether conditioning-prompt ports across models.

~2 days

Thank you

Questions & Discussion

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